

UNITED STATES PATENT APPLICATION

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TITLE OF INVENTION

TECHNICAL FIELD

The present invention relates to content quality assurance systems for autonomous artificial intelligence content generation platforms, and more particularly to a quality gate system that generates multiple content variants per evaluation round, scores each variant against a calibrated multi-dimensional rubric anchored to a known reference model's quality baseline, retries failed evaluations with rotated optimization objectives, adapts the acceptance threshold based on real-world engagement feedback, and integrates training infrastructure readiness signals to provide contextual explanations when quality ceilings are encountered.

BACKGROUND OF THE INVENTION

Autonomous AI content generation systems for social media face a fundamental quality control challenge: generated content must meet a minimum quality standard before publication, but regenerating content until it meets a fixed threshold is expensive in time and compute. A binary retry loop that regenerates content until it passes is unbounded in its retry count. A single-attempt system that publishes whatever is generated provides no quality floor. Neither approach balances throughput and quality effectively.

A further challenge is calibrating the quality threshold against a known reference point. Setting an arbitrary threshold (e.g., "score at least 70 out of 100") provides no meaningful anchor. A threshold calibrated to the quality of a state-of-the-art reference model (e.g., "at least 90% of the quality achieved by Google's Veo model on this rubric") provides a meaningful and defensible floor that can be updated as the reference model improves.

A third challenge is adapting the threshold dynamically based on real-world outcomes: a threshold that generates content that

consistently overperforms on engagement should be raised (the content is better than needed and quality should be improved); a threshold that generates content that consistently underperforms should be lowered (the quality bar is too aggressive and good-enough content is being rejected).

A fourth challenge is explaining quality gate failures in the context of training infrastructure readiness: if the model is in a cold-start state with no synced weights and no calibration, a quality gate failure has a different cause and remedy than a failure when the system is fully calibrated and optimally trained.

There exists a need for a quality gate system that (1) generates large batches of variants per evaluation round, (2) anchors the acceptance threshold to a calibrated reference model quality baseline, (3) rotates the optimization objective on each retry round to explore different content generation strategies, (4) adapts the threshold based on real engagement outcomes, and (5) provides training readiness context with each failure.

SUMMARY OF THE INVENTION

The present invention provides a content quality gate system comprising a multi-round retry loop that generates configurable numbers of content variants per round, a Veo-score-calibrated acceptance threshold derived from the quality output of a reference generative model, an objective rotation mechanism that tries a different optimization angle on each retry round, a pressure-adjusted floor that prevents rejection under extreme deadline pressure, a per-user adaptive threshold stored in a persistent key-value cache, and a training infrastructure readiness profile that contextualizes quality gate outcomes.

In a first aspect, the invention provides a method of quality-gating generated content comprising: generating a first batch of content variants using a primary generation strategy;

scoring each variant against a multi-dimensional rubric; returning the best-scoring variant if its score meets or exceeds a calibrated threshold; and, if no variant meets the threshold, repeating with a rotated optimization objective and an enlarged variant batch until a maximum number of rounds is exhausted, at which point the best variant found is accepted if it meets an absolute pressure floor and rejected otherwise.

In a second aspect, the invention provides a threshold calibration system that derives the acceptance threshold as a fixed percentage of the average score achieved by a reference generative model on the same rubric, and updates this threshold dynamically every six hours by querying the training infrastructure for current model state signals.

In a third aspect, the invention provides an engagement-based threshold adaptation system that raises the per-user acceptance threshold by one point when published content achieves engagement above a platform-specific high-engagement benchmark, and lowers it by one point when engagement falls below a platform-specific low-engagement benchmark.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

I. Quality Gate Architecture

The content quality gate sits between the content generator and the content scheduler. Its inputs are: a user identifier, a base content context (platform, objective, business vertical, brand voice), and an optional override threshold. Its output is a `QualityGateResult` containing the winning variant, the round it passed on, the total number of variants tried, the list of rejected variants, the threshold used, and a storage key for the archived session.

II. Veo-Score Calibration

The acceptance threshold is derived from the empirical quality level of Google's Veo generative video model as measured by this pipeline's multi-dimensional scoring rubric. Veo consistently scores approximately 90 to 95 on this rubric. The design requirement is that Max Booster content should achieve at least 90% of Veo's quality level:

$$\text{DEFAULT_THRESHOLD} = 90\% \times 90 = 81$$

The absolute pressure floor – the minimum score below which content will never be published regardless of deadline pressure – is:

$$\text{VEO_PRESSURE_FLOOR} = 73 \text{ (approximately 90\% of the } \text{DEFAULT_THRESHOLD})$$

These thresholds are not static: the MaxCore Score Calibrator updates them every six hours by querying the training infrastructure for current model capability signals. As the underlying model accumulates more training experience, the calibrated thresholds may rise above the static defaults.

III. Multi-Round A/B Retry Loop

The retry loop proceeds as follows:

Round 1: The Advanced Social AI generator produces `VARIANTS_PER_ROUND` (default 30) variants using the highest-quality semantic generation strategy. If this fails, it falls back to the template generator.

Rounds 2 through `MAX_ROUNDS` (default 10): The template or Python AI generator produces `VARIANTS_PER_ROUND + round` variants (increasing with each round) with a rotated optimization objective.

The objective rotation follows a fixed sequence:

Round 2: viral

Round 3: awareness

Round 4: conversions

Round 5: engagement

Round 6: viral

(pattern repeats)

This rotation ensures that each retry explores a fundamentally different content generation strategy, rather than regenerating semantically identical content.

After each round:

- If the best variant in the batch scores $\geq$ threshold: it is the winner; the loop exits
- If no variant meets the threshold: all variants are added to the rejected list and the next round begins

After exhausting all MAX_ROUNDS rounds:

- The best variant across all attempted variants is identified
- If its score $\geq$ VEO_PRESSURE_FLOOR: it is accepted as the best-available winner (noted in logs)
- If its score $<$ VEO_PRESSURE_FLOOR: the gate returns null; the caller must skip publication

IV. Variant Batch Sizing

The number of variants generated per round increases with the round number:

Round r : $n_variants = VARIANTS_PER_ROUND + r$

This means Round 1 generates 31 variants, Round 2 generates 32 variants, ..., Round 10 generates 40 variants. The increasing batch size reflects the observation that later rounds have already failed with smaller batches, and the probability of finding a passing variant increases with batch size.

V. Training Infrastructure Readiness Awareness

Before each gate run, the system queries the training infrastructure readiness profile, which classifies the system into one of four levels:

- cold: No MaxCore base weights synced and calibration not run
- warming: Some weights synced or calibration pending
- ready: All weights synced, calibrated at defaults
- optimal: All weights synced, training-calibrated above defaults

When the system is cold or warming, calibration is triggered non-blockingly so that subsequent gate runs benefit from calibrated thresholds. Each gate failure includes a readiness hint explaining why the quality ceiling may be lower than expected and what will improve it.

VI. Multi-Dimensional Scoring Rubric

Content variants are scored on ten dimensions:

- engagement: predicted engagement rate (weight 0.25)
- hookStrength: first-sentence attention capture (weight 0.18)
- callToActionEffectiveness: actionability of the post's ask (weight 0.13)
- sentiment: emotional resonance and positivity (weight 0.10)
- clarity: readability and message clarity (weight 0.08)
- brandAlignment: consistency with brand voice profile (weight 0.08)
- algorithmAlignment: platform algorithm optimization signals (weight 0.08)
- specificity: concreteness of claims and data points (weight 0.05)
- emotionalArc: narrative arc from hook to CTA (weight 0.03)
- narrativeAuthenticity: domain-specific vocabulary authenticity (weight 0.02)

The overall score is the weighted sum of the ten dimension scores, each in $[0, 100]$. Weights are calibrated by the MaxCore Score Calibrator every six hours using current model signals from the training infrastructure.

VII. Platform Constraints

Before scoring, each variant is checked against platform-specific constraints including character limits, hashtag count limits, link placement rules, and emoji density norms. Platform constraint violations are penalized in the scoring rubric. The platform-optimized version of the content is used for the constraint check, enabling the scorer to evaluate the version that would actually be published.

VIII. Pressure-Adjusted Quality Gate

The content quality pipeline integrates Caffeine Mode pressure into the scoring gate:

```
pressureAdjustedMinScore(baseMin):  
if pressure = 0: return baseMin  
if pressure > 1.5: return max(VEO_PRESSURE_FLOOR, baseMin - 10)  
if pressure > 0.5: return max(VEO_PRESSURE_FLOOR, baseMin - 7)  
else: return max(VEO_PRESSURE_FLOOR, baseMin - 4)
```

Simultaneously, urgency-themed content (posts that signal time-sensitivity, limited availability, or deadline relevance) receives bonus points in the engagement and hookStrength dimensions proportional to the current pressure. This means that under pressure, the gate lowers but the type of content that can clear the lowered gate is specifically urgency-themed content – maintaining the quality-of-type constraint even while relaxing the quality-of-score constraint.

IX. Per-User Adaptive Threshold

The quality gate retrieves the per-user acceptance threshold from a two-tier lookup:

Tier 1 (PDIM cache): A persistent key-value store keyed as `mbs:quality:threshold:{userId}` with a 30-day TTL. This stores the live adaptive threshold updated by engagement feedback.

Tier 2 (Database fallback): The user's configured threshold from the autopilot preferences table, bounded below by the calibrated `DEFAULT_THRESHOLD`.

After each published post's engagement outcome is observed, the threshold is adapted:

```
if engagementRate ≥ platform_high_benchmark: threshold ←  
  min(95, threshold + 1)  
if engagementRate < platform_low_benchmark: threshold ←  
  max(VEO_PRESSURE_FLOOR, threshold - 1)
```

Platform-specific engagement benchmarks:

```
Twitter/X: high=2.0%, low=0.5%  
Instagram: high=5.0%, low=1.0%  
TikTok: high=8.0%, low=3.0%  
LinkedIn: high=4.0%, low=1.0%  
Facebook: high=2.0%, low=0.5%  
Threads: high=3.0%, low=1.0%  
YouTube: high=4.0%, low=1.0%
```

X. Trained Model Fast Path

When the training infrastructure is in a ready or optimal state, content may be generated directly by the trained model rather than the template/AI pipeline. In this case, the gate applies a `scoreAndGateExisting()` fast path: the trained model's output is scored directly, and if it passes the threshold it is returned immediately without any retry loop. If it fails, the full A/B retry gate is triggered as a fallback. This fast path eliminates

unnecessary regeneration cycles when the trained model is producing high-quality output.

XI. Session Archival

Every quality gate session – pass or fail – is archived to a persistent content storage namespace with: the user identifier, timestamp, threshold used, the round that produced the winner, total variants tried, winner metadata (headline, scores), and rejected variant metadata. This archive feeds back into MaxCore training as a training feedback signal, providing signal about what content quality looks like across the full distribution of attempted outputs.

CLAIMS

1. A computer-implemented system for quality-gating artificially generated content before publication, the system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: maintain a per-user acceptance threshold derived from a fixed percentage of the quality level achieved by a reference generative model on a multi-dimensional scoring rubric; execute a multi-round retry loop that, in each round, generates a batch of content variants, scores each variant, and accepts the highest-scoring variant if its score meets or exceeds the acceptance threshold; rotate the optimization objective on each retry round to encourage exploration of distinct content generation strategies; increase the batch size on each retry round beyond a base batch size; if no variant meets the threshold after a maximum number of rounds, accept the best variant found if its score meets or exceeds an absolute pressure floor and reject the run otherwise; and adapt the per-user acceptance threshold upward or downward by one point based on whether observed real-world engagement for published content exceeds or falls below a platform-specific engagement benchmark.
2. The system of claim 1, wherein the acceptance threshold is derived as 90 percent of the average score achieved by a reference generative model on the multi-dimensional scoring rubric, and wherein the absolute pressure floor is derived as 90 percent of the acceptance threshold.
3. The system of claim 1, wherein the batch size for round r equals a base batch size plus r , producing an increasing number of variants with each successive retry round.

4. The system of claim 1, wherein the optimization objective rotation follows a fixed sequence comprising at least the objectives of engagement optimization, viral content optimization, brand awareness optimization, and conversion optimization, applied cyclically across successive retry rounds.
5. The system of claim 1, wherein a first retry round applies a primary semantic generation strategy and subsequent retry rounds apply a secondary template-based or AI-based generation strategy, enabling the gate to exploit the highest-quality generation capability on the first attempt before falling back to broader search strategies.
6. The system of claim 1, wherein the per-user acceptance threshold is retrieved from a persistent key-value cache with a configurable time-to-live, and wherein the database stored user preference is used as a fallback when the cache entry is absent or expired.
7. The system of claim 1, wherein the platform-specific engagement benchmarks differ by platform, with platforms characterized by algorithmically amplified content having higher high-engagement thresholds and lower low-engagement thresholds than platforms characterized by follower-based content distribution.
8. The system of claim 1, further comprising a training infrastructure readiness profiler that classifies the training system into one of a plurality of readiness levels and includes a human-readable readiness hint in each quality gate failure response, explaining the relationship between the current readiness level and the observed quality ceiling.
9. The system of claim 8, wherein when the training infrastructure is in a cold or warming readiness level,

the readiness profiler triggers a non-blocking calibration procedure that updates the acceptance threshold and scoring weights from current training infrastructure signals for subsequent gate runs.

10. The system of claim 1, further comprising a scored-and-gate fast path that scores already-generated content from a trained model against the acceptance threshold without generating new variants, and triggers the full multi-round retry loop only when the trained model's content fails the fast path.
11. The system of claim 1, wherein the multi-dimensional scoring rubric comprises at least ten scoring dimensions including engagement potential, hook strength, call-to-action effectiveness, sentiment, clarity, brand alignment, algorithm alignment, specificity, emotional arc, and narrative authenticity, each weighted according to calibrated weights derived from training infrastructure signals.
12. The system of claim 11, wherein the calibrated scoring weights are updated at a configurable refresh interval by querying the training infrastructure for current model capability signals, and wherein default weights are used when the training infrastructure is unavailable.
13. The system of claim 1, further comprising a pressure adjustment mechanism that reduces the acceptance threshold by an amount proportional to a real-time schedule pressure score, subject to a minimum bound of the absolute pressure floor, and simultaneously increases the score contribution of urgency-themed content signals proportional to the same pressure score.
14. A method of generating and quality-gating content for autonomous social media publication, the method comprising:

establishing an acceptance threshold for a user calibrated to a fixed percentage of a reference model's quality on a multi-dimensional rubric; generating a first batch of content variants using a primary generation strategy; scoring each variant and accepting the best-scoring variant if its score meets the acceptance threshold; if no variant meets the threshold, generating successive batches with rotated optimization objectives and increasing batch sizes until either a variant meets the threshold or a maximum retry count is reached; if the maximum retry count is reached, accepting the best variant found if it meets an absolute pressure floor, else rejecting the run; publishing the accepted variant; and observing the published content's real-world engagement rate and adapting the user's acceptance threshold upward if engagement exceeds a platform-specific high benchmark or downward if engagement falls below a platform-specific low benchmark.

15. The method of claim 14, wherein adapting the threshold is bounded below by the absolute pressure floor and bounded above by a maximum threshold value.
16. The method of claim 14, further comprising archiving metadata for every quality gate session, including the threshold used, the number of rounds attempted, the scores of the winning and all rejected variants, and using the archived data as training feedback for an underlying generative model.
17. The method of claim 14, further comprising pushing a structured training feedback signal to a connected training infrastructure after each engagement observation, the signal including the platform, content type, engagement rate, and a curriculum hint identifying whether the signal represents a case for reinforcing a winning pattern or improving a weak pattern.

- 18.** A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement: a variant batch generator configured to produce a configurable number of content variants per retry round and increase the batch size with each successive round; a multi-dimensional scorer configured to evaluate each variant on a plurality of weighted dimensions and compute an overall score as a weighted sum; an objective rotator configured to supply a different optimization objective string on each retry round from a fixed rotation sequence; a threshold manager configured to retrieve a per-user acceptance threshold from a persistent cache, fall back to a database preference, and persist updated thresholds after engagement observations; a readiness profiler configured to classify the training infrastructure into a plurality of readiness levels and expose a readiness hint with each gate outcome; and a session archiver configured to record the complete outcome of each gate run, including winner and rejected variant metadata, to a persistent storage namespace.
- 19.** The computer-readable medium of claim 18, further storing instructions that implement a scored-and-gate method that accepts an externally generated content string, scores it directly, and returns it immediately if it passes the threshold without generating any new variants.
- 20.** The computer-readable medium of claim 18, wherein the threshold manager enforces that the per-user acceptance threshold may not fall below the absolute pressure floor regardless of how many consecutive low-engagement outcomes are observed.

ABSTRACT

A content quality gate generates multiple variants of AI-produced social media content per evaluation round, scores each on a ten-dimension rubric calibrated so that a default acceptance threshold of 81/100 represents 90 percent of the quality achieved by Google's Veo model on the same rubric. Content that meets the threshold is accepted immediately. Content that fails triggers a retry with a rotated optimization objective and an enlarged variant batch, up to ten rounds. After ten rounds, the best variant found is accepted if it scores at least 73/100 (the absolute pressure floor); otherwise the run is rejected. A per-user threshold is stored in a persistent cache and adapts one point upward when published content over-performs relative to platform-specific engagement benchmarks, and one point downward when it under-performs. A training infrastructure readiness profiler classifies system readiness into four levels and includes contextual remediation hints with each gate failure. A pressure adjustment mechanism reduces the threshold proportionally to real-time schedule deficit pressure, while simultaneously increasing the score of urgency-themed content, maintaining the quality-of-type constraint under deadline pressure.